

Decision Tree Classifier Using Gini Impurity and Information Gain

Project Overview :-

1. Creates a stock market decision dataset.
2. Calculates Gini Impurity for the target variable.
3. Calculates Information Gain for each feature.
4. Identifies the best feature for the root node.
5. Encodes categorical variables using One-Hot Encoding.
6. Trains a Decision Tree Classifier.
7. Visualizes the generated Decision Tree.

Dataset :--

Feature	Description
MarketTrend	Market direction (Up/Down)
Volume	Trading volume (High/Low)
News	Market news sentiment (Positive/Negative)
RSI	Relative Strength Index (High/Medium/Low)
Decision	Target variable (Buy/Sell)

Technologies Used :-

- Python
- Pandas
- Scikit-Learn
- Matplotlib

Conclusion:

This project shows how a Decision Tree uses Gini Impurity and Information Gain to make decisions and classify data. It also demonstrates training and visualizing a Decision Tree using Python and Scikit-Learn.